Data Science

CSE558

Recap

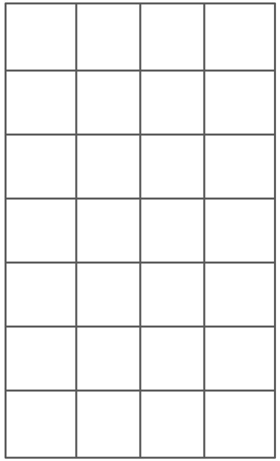
- Estimating Distinct Elements

Overview

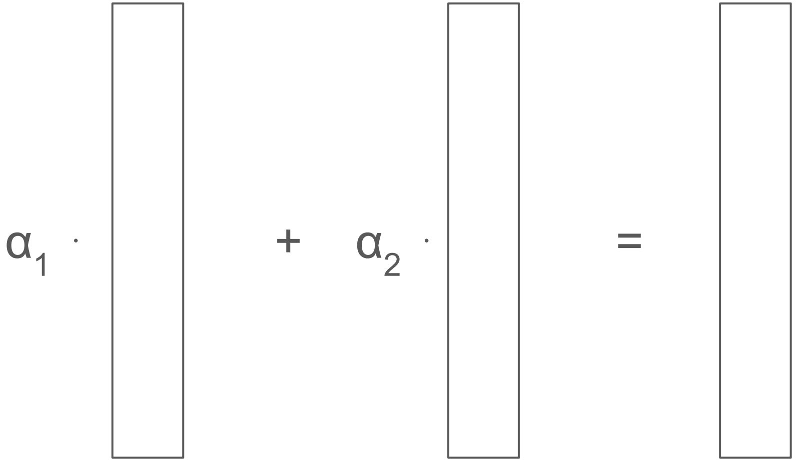
- High dimensional data

- Dimensionality Reduction

Linear Algebra

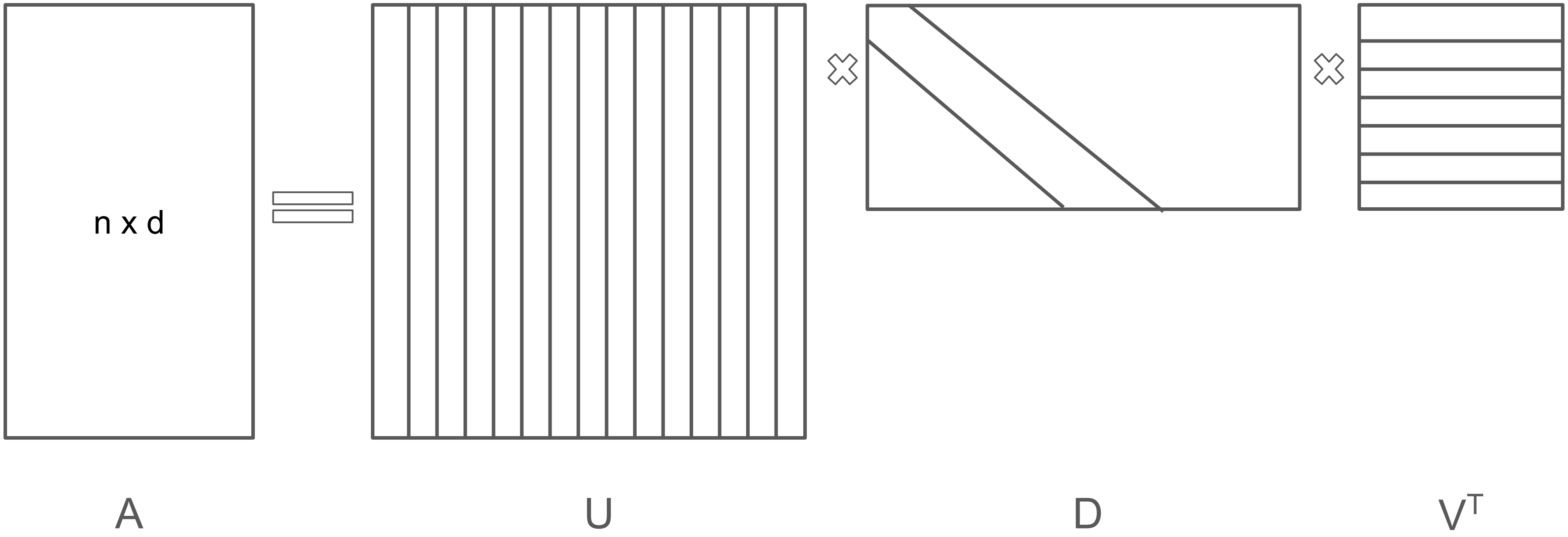


Rank-2

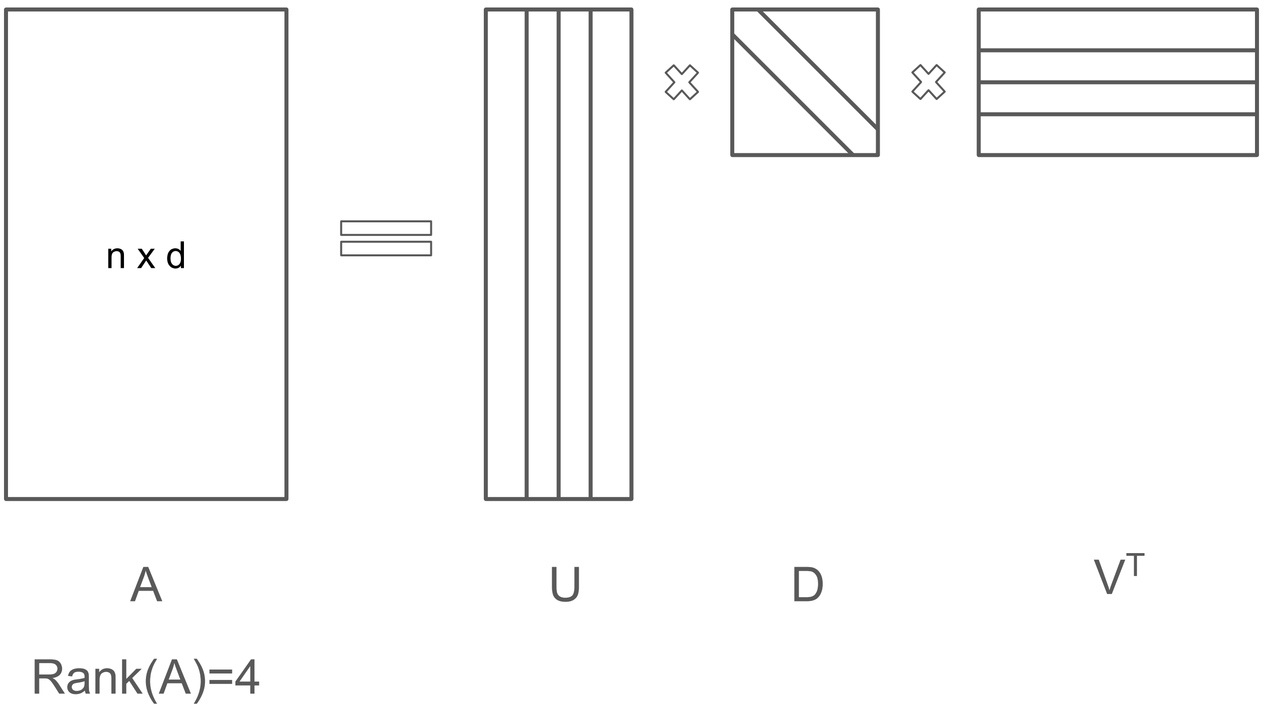


Rank: Maximal number of linearly independent columns or rows

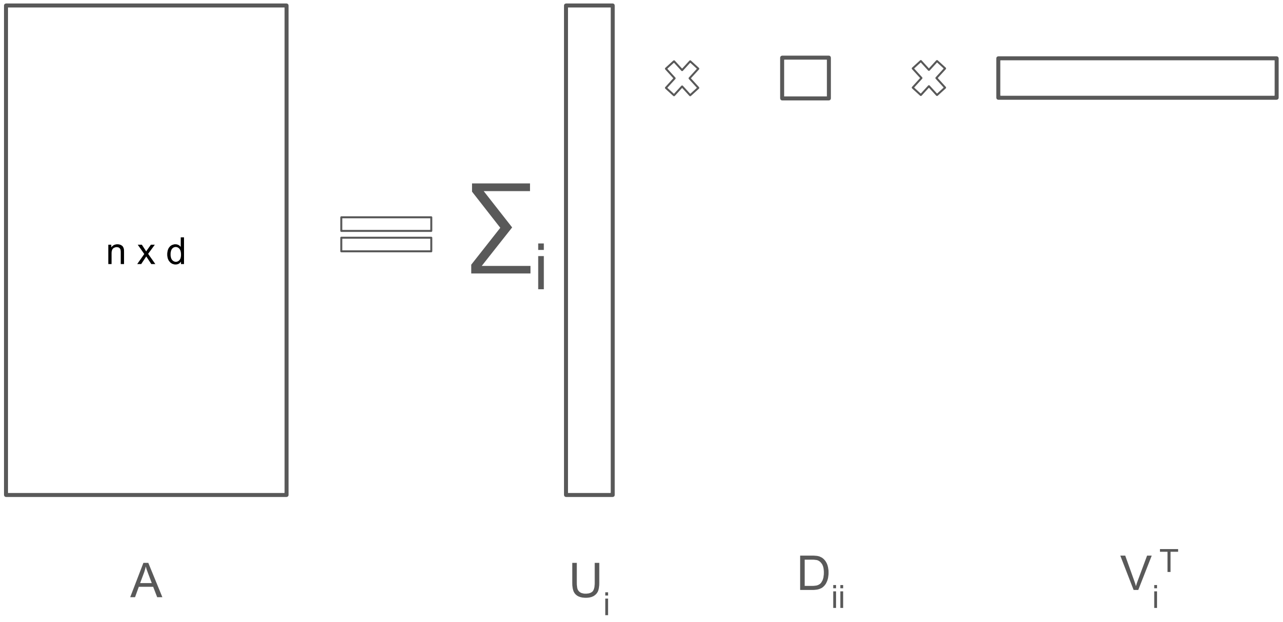
Singular Value Decomposition



Singular Value Decomposition



Singular Value Decomposition



Singular Vectors and Singular Values

$$\mathbf{v}_1 \in \underset{\|\mathbf{v}\|_2=1}{\operatorname{argmax}} \|A\mathbf{v}\|_2 \qquad \sigma_1(A) = \|A\mathbf{v}_1\|_2$$

$$\mathbf{v}_2 \in \underset{\mathbf{v}\perp\mathbf{v}_1;\|\mathbf{v}\|_2=1}{\operatorname{argmax}} \|A\mathbf{v}\|_2 \qquad \sigma_2(A) = \|A\mathbf{v}_2\|_2$$

Singular Vectors and Singular Values

Running time $O(nd \cdot \min(n,d))$

Compute, $B = A^TA$ and then compute eigen value decomposition of B.

<u>Power Iteration:</u>

Sample a random vector x_0 and repeat following for k steps

$$\mathbf{x}_{t+1} = \frac{A\mathbf{x}_t}{\|A\mathbf{x}_t\|_2}$$

How to compute the next pair?

Projection

Projection of point **a** on a line passing through point **x** is $\left(\frac{\mathbf{x}\mathbf{x}^T}{\|\mathbf{x}\|_2^2}\right)\mathbf{a}$

Orthogonal distance $\left\|\left(\mathbf{I}-\frac{\mathbf{x}\mathbf{x}^T}{\|\mathbf{x}\|_2^2}\right)(\mathbf{a}-\mathbf{b})\right\|_2^2$

Questions

Let A be an n by d matrix with singular vectors $v_1, v_2, \ldots, v_r$. For $1 \le k \le r$, let V_k be the subspace spanned by $v_1, v_2, \ldots, v_k$. For each k, V_k is the best k-dimensional subspace for A.

Let, $A_k = \sum_{i=1}^{k} \sigma_i \mathbf{u_i} \mathbf{v_i}^T$ then for any B,

$$\|A - A_k\|_F \le \|A - B\|_F$$

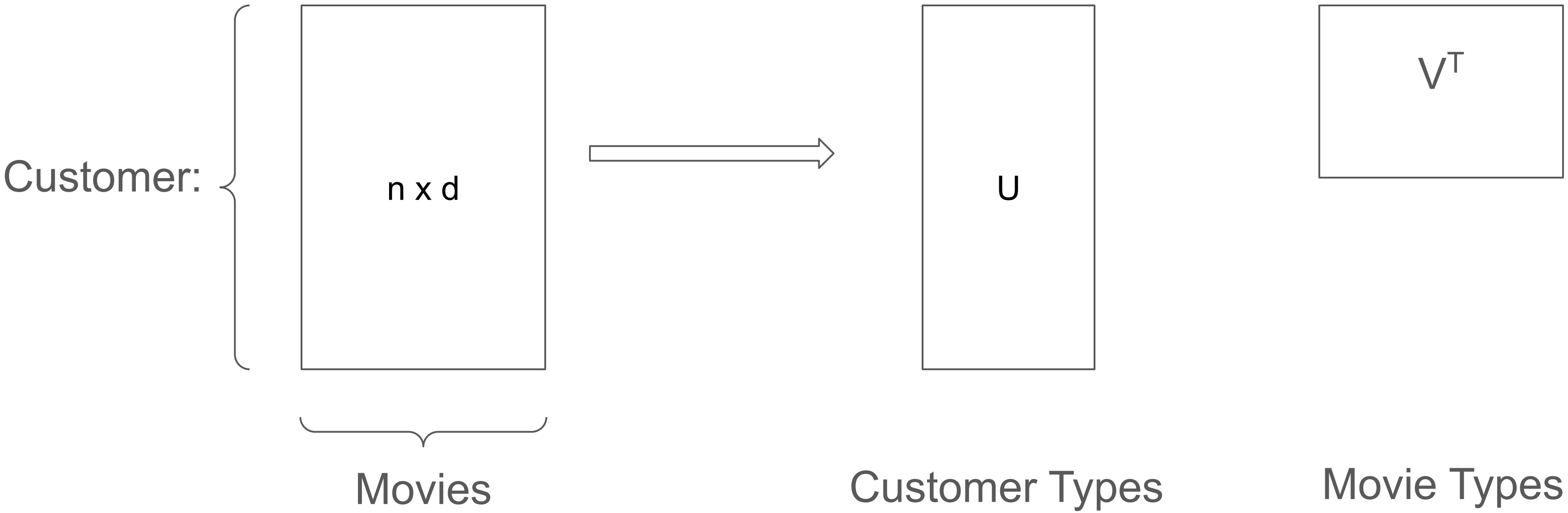
$$|A\mathbf{w_1}|^2 + |A\mathbf{w_2}|^2 \le |A\mathbf{v_1}|^2 + |A\mathbf{v_2}|^2$$

Overview

- High dimensional data
- Dimensionality Reduction

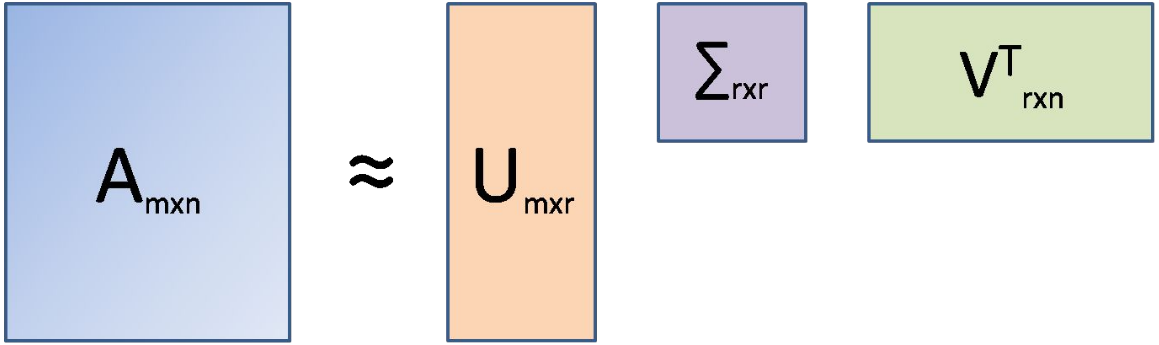
Dimensionality Reduction

Recommendation System (k types of movies):



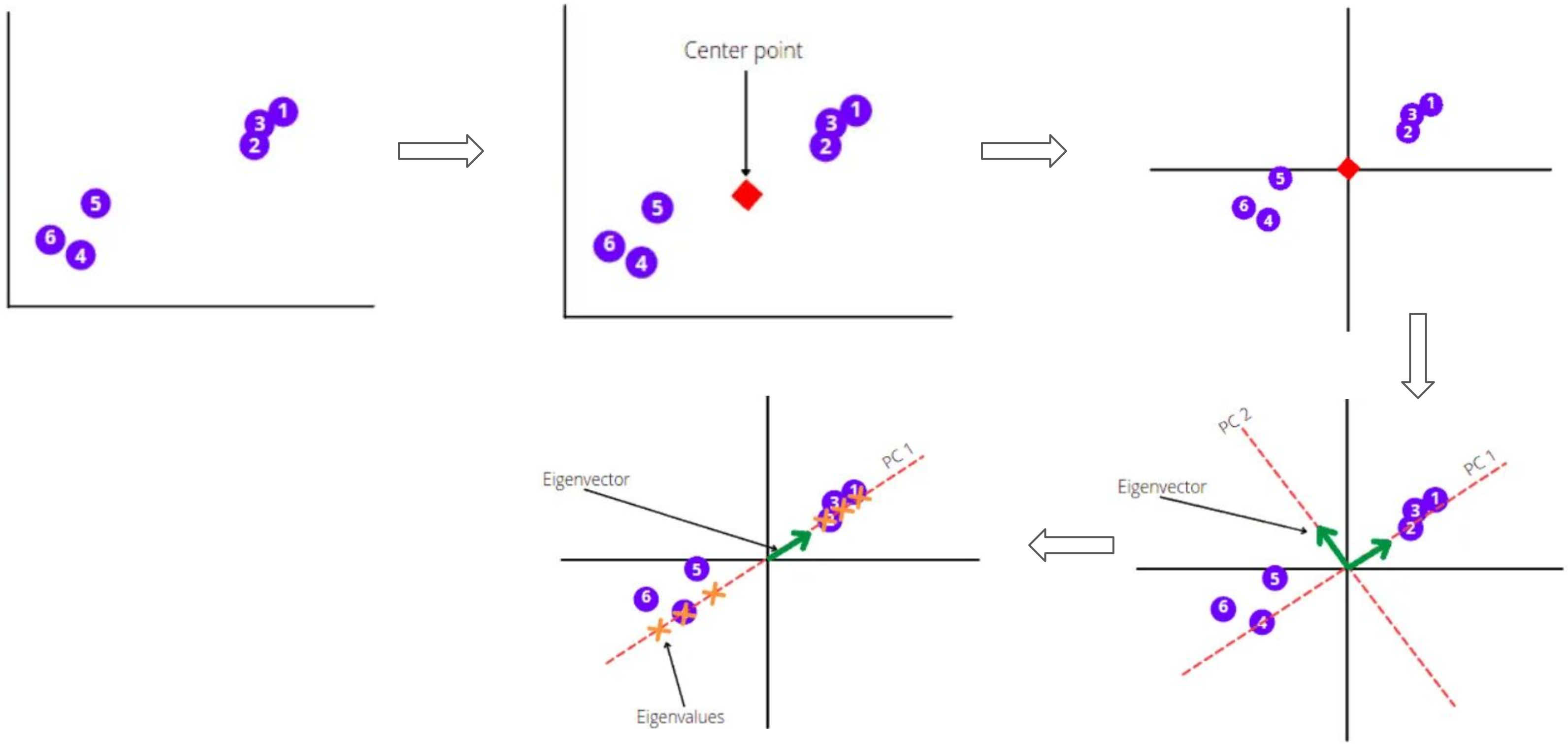
Dimensionality Reduction

Latent semantic analysis



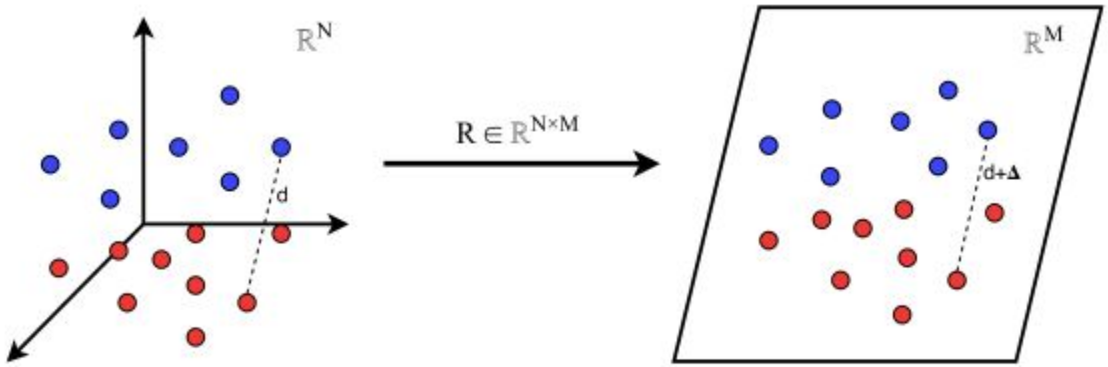
Application of SVD

PCA



Dimension Reduction

Projection



Simple Case

For any set of n points (x1, x2, …, xn) in a d-dimension. The goal is to preserve the following for every pair of points (x, y) and for some ε ∈ (0,1).

$$(1 - \epsilon) // xi - xj // \le // \tilde{x}i - \tilde{x}j // \le (1 + \epsilon) // xi - xj //$$

- Consider that the points are on 1st axis.
- Consider that the points are on a straight line.
- Consider that the points are on a plane (or 2-dim space).
- Consider that the points are on a k-dim space.